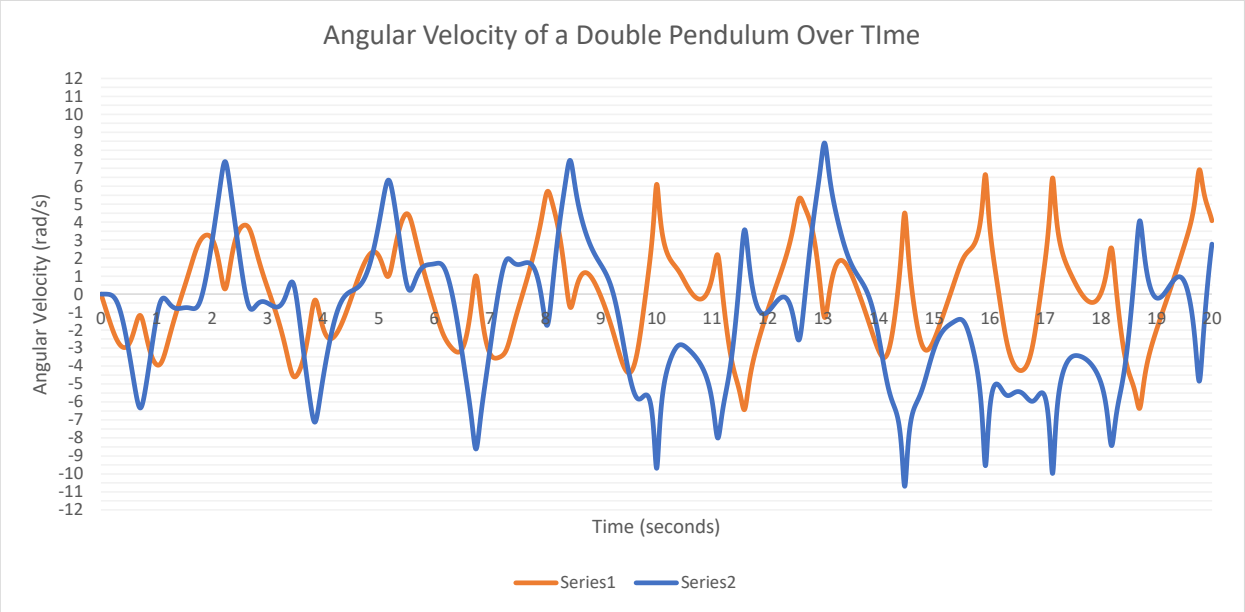
Double Pendulum

Simulation & Data Analysis | 20-second simulation

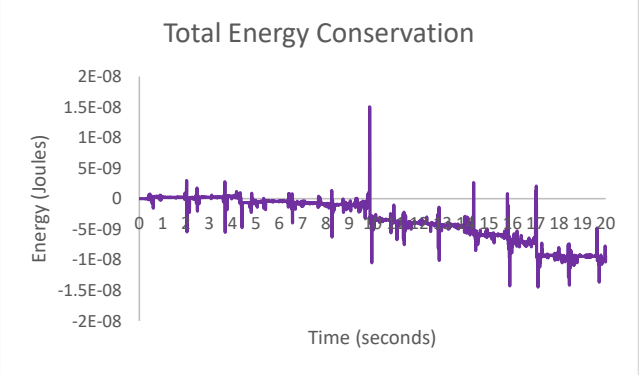
OBSERVATIONS	ENERGY RANGE
5,000	2.95E-08
Data points recorded	Joules

MAX SPEED PENDULUM 1	MAX SPEED PENDULUM 2
6.92	10.69
rad/s	rad/s

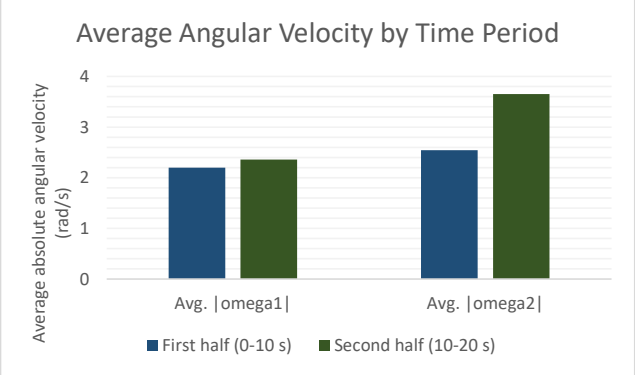
01 Angular Velocity Over Time



02 Energy Conservation



03 First Half vs Second Half



KEY FINDINGS

- Both pendulums exhibit changing angular velocities throughout the simulation.
- Pendulum 2 reaches a higher maximum absolute angular velocity than pendulum 1.
- Total mechanical energy remains close to its initial value, with a very small numerical deviation.
- Average absolute angular velocity increases in the second half of this particular simulation, especially for pendulum 2.
- The energy range is approximately 2.95*10^-8, supporting the conclusion that the numerical simulation approximately conserves total mechanical energy.